Progress & Priorities

U.S. particle physics advances the frontiers of discovery science, builds and trains a highly skilled technical workforce, and advances U.S. leadership worldwide.

The 2023 P5 report provides the long-term strategy and priorities for U.S. investments in particle physics for the next 10 years within a 20-year vision.

The top priorities in 2025 are to advance current construction projects, support operations of ongoing experiments, and strengthen support for research at universities and national laboratories to maximize the opportunities for scientific discoveries.

Support for big data analysis, innovative instrumentation, a vibrant theory program, and future accelerator development is essential to maintain U.S. leadership and train the next generation of scientists. U.S. particle physics also plays a key role in national initiatives in microelectronics, Artificial Intelligence and Machine Learning, and quantum information science.

It is critical to complete construction and deliver science from the following projects:

High-Luminosity Large Hadron Collider (HL-LHC) is an accelerator and detector upgrade to continue the highly successful LHC program and bilateral partnership with CERN. This will allow for precise measurements of the Higgs boson and its interactions. Searches for new physics using state-of-the-art machine learning techniques at the highest-ever center-of-mass energy will broaden our understanding of fundamental physics.

Deep Underground Neutrino Experiment (DUNE) is the first U.S.-hosted world-leading international science mega-project. It will determine the fundamental properties of the mysterious neutrinos that permeate the cosmos. DUNE will be powered by the superconducting PIP-II linear accelerator at Fermilab.

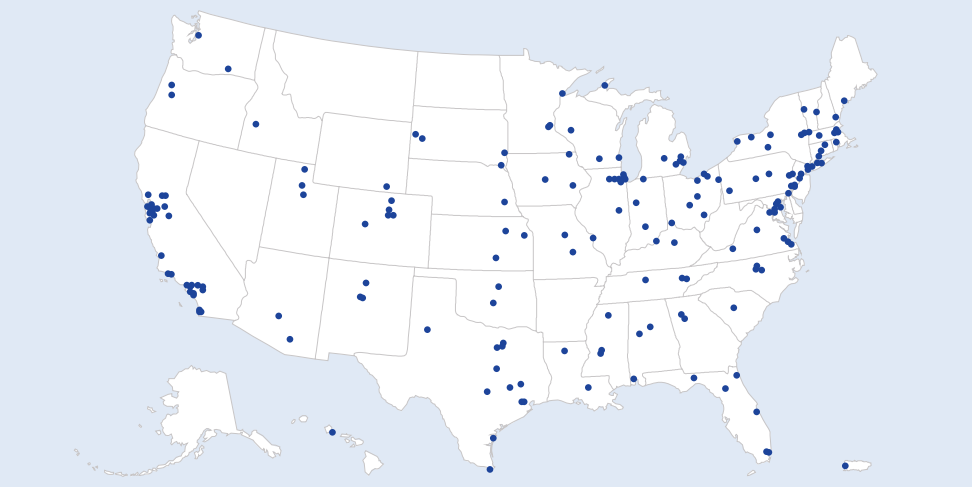
Vera C. Rubin Observatory is a major new telescope and camera nearing completion in Chile. Its Legacy Survey of Space and Time (LSST) will study cosmic evolution and other fundamental astrophysical questions.

CMB-S4 is a next-generation cosmic microwave background (CMB) experiment with an array of telescopes in the Chilean Atacama Desert. It will observe fingerprints from the earliest moments of the Universe to understand its origins.

These carefully chosen investments will enable a steady stream of exciting new results for many years to come and will maintain U.S. leadership in key areas.

Particle Physics in the United States

Particle physics is both global and local. Scientists, engineers, and technicians at over 200 universities, institutes, and laboratories throughout the U.S. are working in partnership with their international colleagues to build high-tech tools and components and conduct scientific research. Particle physics excels at building a highly-skilled technical workforce across the country by training and educating the next generation of innovators. Particle physics activities in the U.S. attract leading scientists from around the world.



The P5 strategy continues to be very successful. The 2023 P5 report builds upon these successes and charts the course to a new era of discovery, innovation, and U.S. leadership for the next decade and beyond.

Recent Science Results

The **ATLAS** and **CMS** collaborations at the **LHC** have each produced more than 1,300 refereed publications as they continue to probe the fundamental properties of the Higgs boson and expand their reach for new particles, forces of nature, and physical principles. Among dozens of examples, they have probed very rare decays of the Higgs boson, observed the simultaneous production of four top quarks, and detected quantum entanglement in top-quark pairs at the highest energies.

The **LHCb** experiment discovered new configurations of quarks, measured differences between the properties of matter and antimatter, and reported evidence for an intriguing deviation from the expectation in the properties of leptons.

The **Dark Energy Survey** collaboration has published cosmological results from supernovae and the baryonic acoustic oscillations using their final data. Together they can set strong constraints on the expansion history of the Universe.

The **Dark Energy Spectroscopic Instrument (DESI)** has mapped over 39 million galaxies and quasars, creating the largest ever 3D map of the Universe. The third-year DESI results, accompanied by a publicly available data release, provide the most precise dark energy measurements ever made and indicate that dark energy may vary over cosmic time.

The **IceCube** experiment in Antarctica used an AI model to detect seven high-energy tau neutrino interactions of astrophysical origin.

The **NOvA** and **T2K** experiments, based at Fermilab and in Japan respectively, have produced their first joint analysis of long-baseline neutrino oscillation. The new results provide insight into the mass ordering and flavor symmetry of neutrinos.

The **COHERENT** collaboration measured for the first time the coherent scattering of neutrinos with germanium.

The **LUX-ZEPLIN (LZ)** dark matter experiment in South Dakota has published new results, providing world-leading constraints on the particle identity of dark matter across a vast range of mass scales, and promising greater sensitivity with continued data collection.

Program Advances and Milestones

The **CMS and ATLAS upgrades**, built at U.S. institutions, have begun arriving at CERN for assembly. The first two of the ten **HL-LHC** focusing magnets are delivered to CERN. They will be used to demonstrate the capabilities of the new superconductor technologies to provide the high luminosity for the HL-LHC and beyond, towards much higher energy future colliders.

The **Long Baseline Neutrino Facility (LBNF)** construction is on schedule. Excavation of the caverns has been completed in South Dakota and work on outfitting the caverns for installation of the Far Detector (FD) has started. Cryostat components continue to be delivered for installation and deployment in the caverns. Fermilab will construct a new beamline and detector hall to host the Near Detector (ND). This new beamline, and the combination of the ND and FD will be essential components of **Deep Underground Neutrino Experiment (DUNE)**, which is slated to take first data at the end of this decade.

The **LSST Camera at the Vera C. Rubin Observatory**, the world's largest camera for astrophysics has been installed in Chile. Commissioning of the three-ton, 3.2-billion-pixel camera begins in summer 2025.

The **Short Baseline Near Detector (SBND)** at Fermilab has been filled with liquid argon and has detected its first neutrinos.

All four **SuperCDMS** towers have been delivered from SLAC to SNOLAB in Canada. Construction is projected to finish in mid-2025, with commissioning through the end of 2025.

Theoretical physicists have discovered connections among quantum entanglement, information, and gravity, shedding important new light on the black hole paradox pointed out by the late Stephen Hawking, and suggesting novel ideas about the emergence of spacetime. Theorists are also active in the use of AI/ML and developing QIS techniques to simulate the quantum world of elementary particles, as well as devising new ways to use quantum sensors in novel searches for dark matter.

Prepared by

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